Year 7
Mathematics

Unit 1: Analysing and Displaying Data

1.5 More Graphs and Tables 2

More Graphs and Tables 2

STARTER · 5 MIN

Date: _____ **Title: More Graphs and Tables 2**

LEARNING OBJECTIVES

- To analyse and present data using spreadsheets in a computer software program

STARTER *Answer in your book. No calculators.*

1. Find the mode: 11, 7, 12, 12, 8, 6, 11, 12, 9, 8
2. Find the median of the same data
3. Find the mean of the same data
4. Find the range of the same data

Starter — Answers

STARTER

Tick or correct in green pen

1. **12**

2. **Median = (9 + 11) ÷ 2 = 10**

3. **Mean = 96 ÷ 10 = 9.6**

4. **Range = 12 − 6 = 6**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can enter data in a spreadsheet and use formulas to find averages	☐
SILVER	I can choose the most appropriate graph to represent data	☐
GOLD	I can draw, read and interpret pie charts and bar-line graphs	☐

Tick each box in the plenary when you can do it on your own.

Introduction — More Graphs and Tables 2

INTRODUCTION · 3 MIN

A spreadsheet can work out averages and draw graphs for you.

Spreadsheet

Rows and columns of cells. B2 is column B, row 2.

Column letter, then row number

Formulas

A formula starts with =

=AVERAGE(A1:A10)
=MEDIAN(A1:A10)

Pie chart

Shows how a whole is split into parts.

The biggest sector is the mode

Bar-line graph

Like a bar chart, but the bars are lines.

Read the top of each line

Examples 1 and 2 — Spreadsheets and Formulas

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint Formulas start with = =AVERAGE( ) finds the mean

Example 1

Look at this part of a spreadsheet.

(a) What is the number in cell B2?

(b) Which cell is the number 81 in?

	A	B
1	81	16
2	54	25

Example 2

Weeks in the Top 10 are in A1 to A10:

14, 10, 12, 16, 10, 11, 11, 13, 12, 10

Write a formula for the mean.

What answer does it give?

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Column letter, then row number

(a) B2 = 25

(b) 81 is in = A1

	A	B
1	81	16
2	54	25

Tip: A1 is the top left cell.

Example 2

Type the formula in cell A11

Formula = =AVERAGE(A1:A10)

Total = 119

Mean = 119 ÷ 10

= 11.9 weeks

Tip: =MODE(A1:A10) gives 10.

Task 1

PRACTICE · 9 MIN

BRONZE

Use the spreadsheet of Lauren's computer game times

Hint Formulas start with =
=AVERAGE( ) finds the mean

Time Lauren played each day

	A	B
1	Day	Time (min)
2	Mon	30
3	Tue	45
4	Wed	40
5	Thu	15
6	Fri	70
7	Sat	120
8	Sun	60

1. What is in cell B3?
2. Which cell holds 120?
3. Write a formula for the mean in B9.
4. Work out the mean to 1 d.p.
5. Work out the median.

★ Work out the range.

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

Time Lauren played each day

	A	B
1	Day	Time (min)
2	Mon	30
3	Tue	45
4	Wed	40
5	Thu	15
6	Fri	70
7	Sat	120
8	Sun	60

1. **45**
2. **B7**
3. **=AVERAGE(B2:B8)**
4. **Mean = 380 ÷ 7 = 54.3 min**
5. **15, 30, 40, 45, 60, 70, 120**
   Median = 45 min

★ **Range = 120 − 15 = 105 min**

Examples 3 and 4 — Choosing the Right Graph

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Measured data: bars touch Change over time: line graph

Example 3

Anita asked 25 girls which instrument they liked most.

Draw a suitable graph.

Instrument	Girls
Clarinet	3
Violin	8
Piano	5
Sax	4
Drum	5

Example 4

Bassi's track lengths (seconds):

50–99: 3, 100–149: 11, 150–199: 6,

200–249: 2, 250–299: 3

Which graph should he draw? Why?

Examples 3 and 4 — Corrections

CORRECTIONS

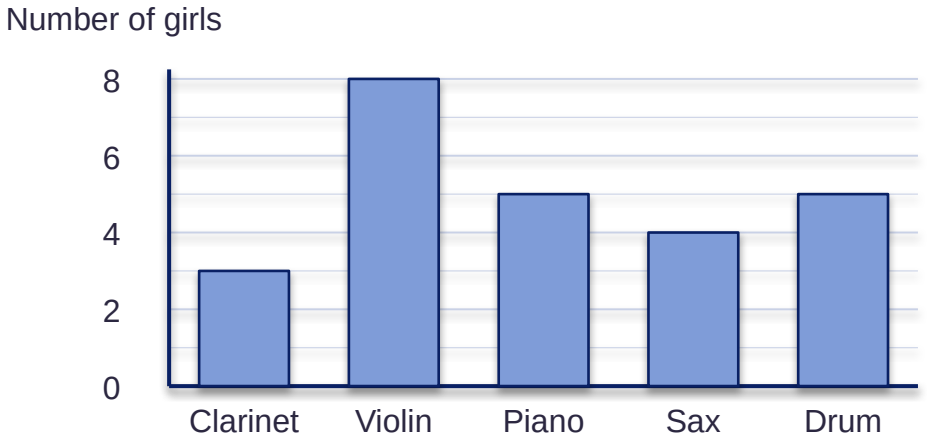
SILVER

Check your copy and correct in green pen

Example 3

Categories: a bar chart

Bars the same width, with gaps

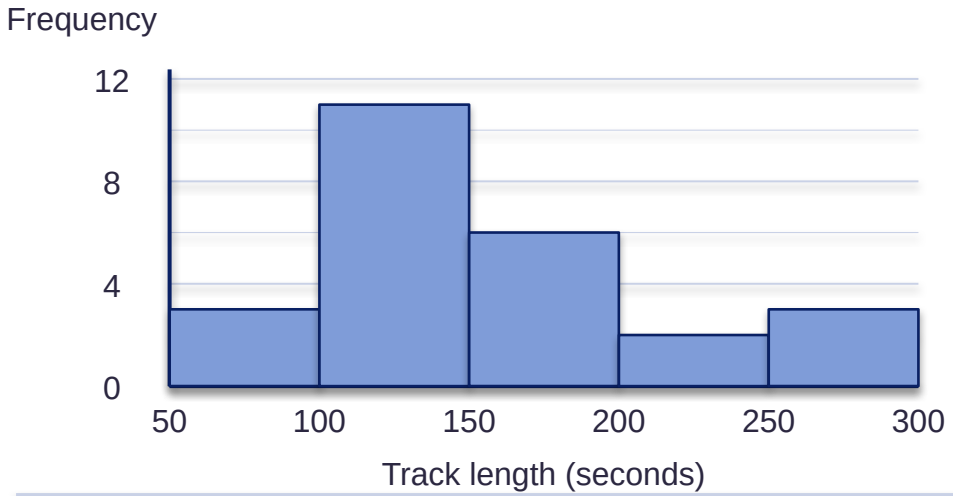


Tip: give the chart a title and label the axes.

Example 4

Time is continuous: bars touch

A frequency diagram



Tip: close the gaps for continuous data.

Task 2

PRACTICE · 9 MIN

SILVER

Choose the best graph for each set of data

Hint Measured data: bars touch
Change over time: line graph

1. The temperature every hour for one day

2. The favourite colours of a class

3. The heights of 30 students

4. How a family spends its money each month

5. Favourite sports of boys and of girls

★ The shoe sizes of 100 people

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. **Line graph:**
   change over time

2. **Bar chart (or pie chart)**

3. **Frequency diagram:**
   height is continuous

4. **Pie chart:**
   parts of a whole

5. **Dual bar chart:**
   compares two groups

★ **Bar-line graph (or bar chart)**

Examples 5 and 6 — Pie Charts and Bar-line Graphs

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

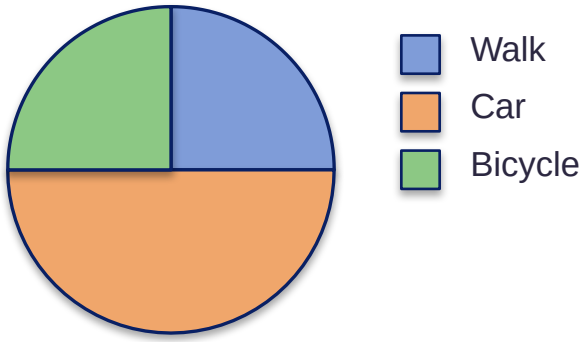
Hint
360° = the whole group
Bar-line: read the top of each line

Example 5

(2 marks)

100 people in Group A said how they travel to work.

Which was the modal method? How many walked?

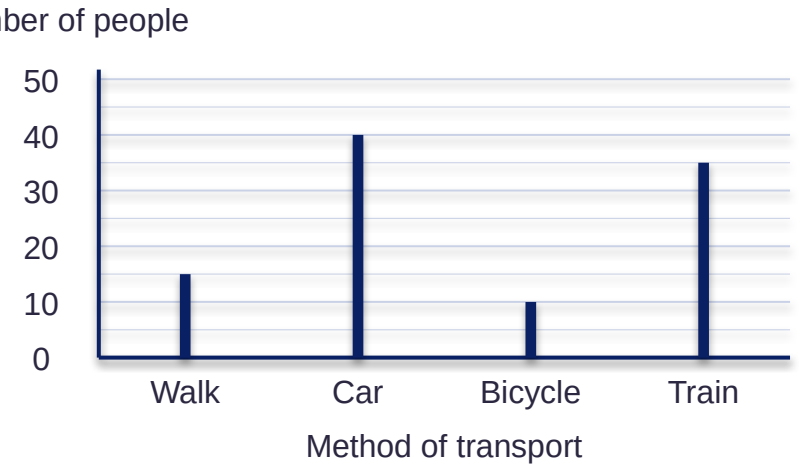


Example 6

(2 marks)

100 people in Group B.

How many more people went by car in Group A than in Group B?



Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

The biggest sector is the mode

Mode = car (half the circle)

Walk is a quarter of the circle.

Walk = 100 ÷ 4

= 25 people

Tip: a quarter of the circle is 90°.

Example 6

Read the top of the Car line

Group A, car = 50

Group B, car = 40

Difference = 50 − 40

= 10 people

Tip: one small square is 5 people.

Task 3

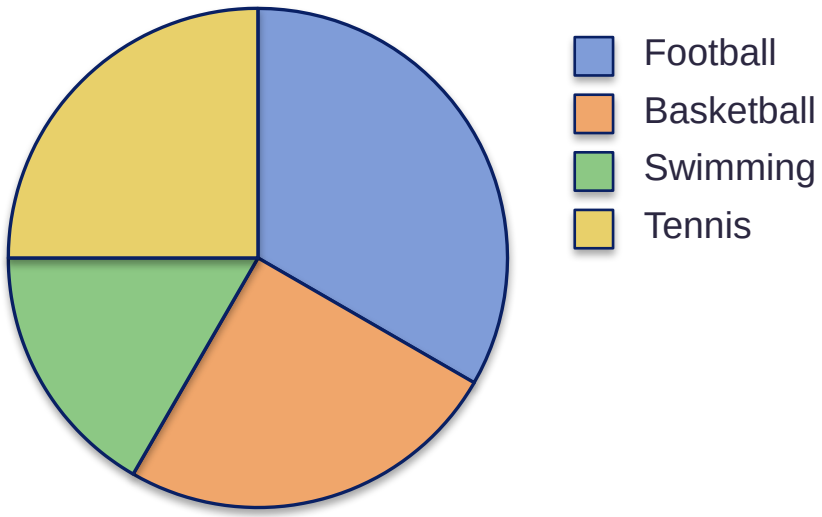
PRACTICE · 9 MIN

GOLD

Use the pie chart: 36 students chose a sport

Hint
360° = the whole group
Bar-line: read the top of each line

Favourite sport of 36 students



1. Which sport is the mode?
2. How many degrees is one student?
3. What is the angle for football?
4. What fraction chose swimming?
5. How many chose tennis?

★ Why can you not find the median here?

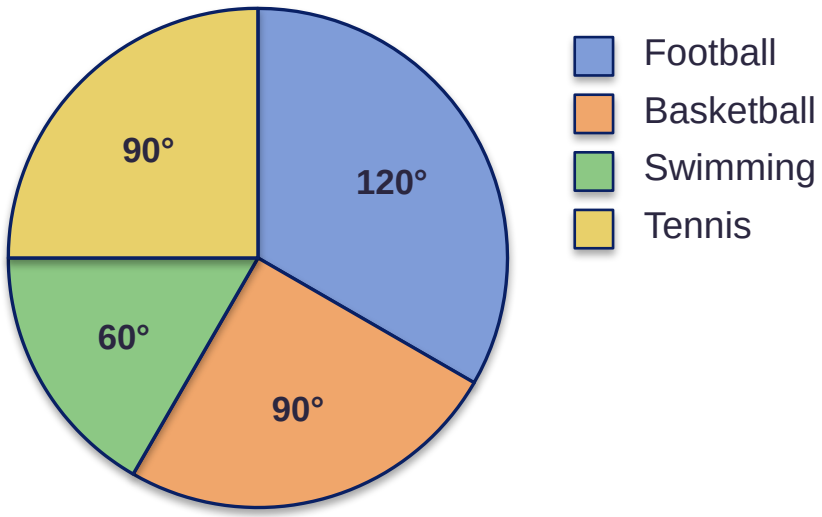
Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

Favourite sport of 36 students



1. **Football**
2. **One student = 360° ÷ 36 = 10°**
3. **Football = 12 × 10° = 120°**
4. **6 out of 36 = $\frac{1}{6}$**
5. **9 students**

★ **The sports are not numbers, so they have no order.**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Write a formula for the mean of cells A1 to A5.

SILVER Which graph shows how a whole is split into parts?

GOLD 40 people: the car sector is 90°. How many went by car?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **=AVERAGE(A1:A5)**

SILVER **A pie chart**

GOLD **Car = 40 ÷ 4 = 10 people**

Exam Practice — Spreadsheets and Bar Charts

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint Formulas start with =
=AVERAGE( ) finds the mean

Question 1

(3 marks)

Part of a spreadsheet.

	A
1	7
2	4
3	9
4	4
5	6

(a) What number is in cell A3? (1)

.....

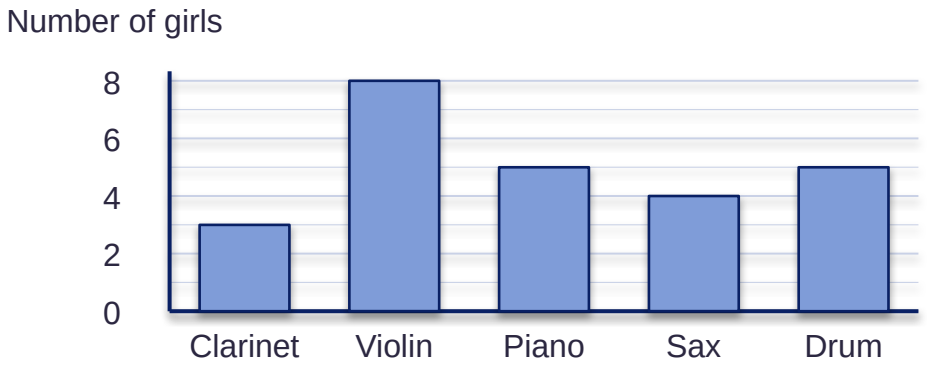
(b) Work out the mean of A1 to A5. (2)

.....

Question 2

(2 marks)

Instruments 25 girls liked most.



(a) How many girls liked the violin? (1)

.....

(b) Which two were equal? (1)

.....

Exam Practice — Choosing Graphs

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Measured data: bars touch
Change over time: line graph

Question 3

(3 marks)

Choose the best graph for each.

(a) The temperature every hour (1)

.....

(b) How a whole budget is split (1)

.....

(c) The masses of 50 apples (1)

.....

Question 4

(2 marks)

Bassi's track lengths.

Length (s)	Frequency
50–99	3
100–149	11
150–199	6
200–249	2
250–299	3

(a) Write down the modal class. (1)

.....

(b) Should the bars touch? Why? (1)

.....

Exam Practice — Pie Charts and Bar-line Graphs

EXAM PRACTICE · 10 MIN

GOLD

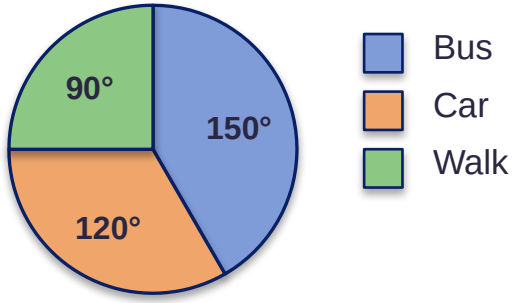
IGCSE exam-style: show all your working

Hint
360° = the whole group
Bar-line: read the top of each line

Question 5

(3 marks)

60 people: how they travel.



(a) Which method is the mode? (1)

.....

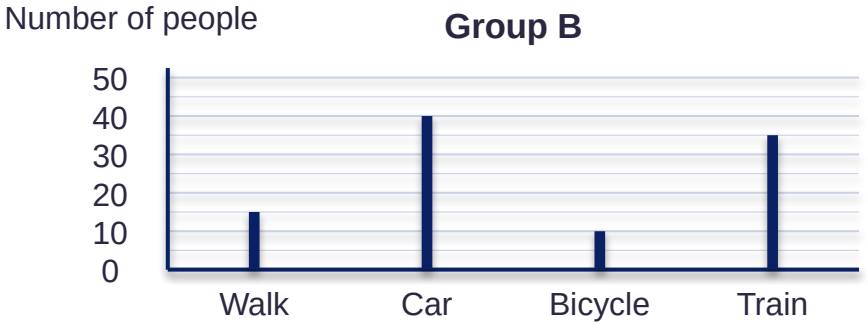
(b) How many people walked? (2)

.....

Question 6

(2 marks)

Group A: 100 people, half by car.



(a) How many in B went by train? (1)

.....

(b) How many more by car in A? (1)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 9 — B1

(b) Total = 7 + 4 + 9 + 4 + 6 — M1

Mean = 30 ÷ 5

= 6 — A1

Tip: =AVERAGE(A1:A5) gives 6.

Question 2

(a) 8 — B1

(b) Piano and drum (5 each) — B1

Tip: read the top of each bar.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) Line graph B1

(b) Pie chart B1

(c) Frequency diagram B1

Tip: think what the data is, then choose.

Question 4

(a) 100–149 B1

(b) Yes: time is continuous B1

Tip: measured data is continuous.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) Bus (150°) — B1

(b) One person = 360° ÷ 60 = 6° — M1

Walk = 90° ÷ 6°

= 15 people — A1

Tip: 90° is a quarter: 60 ÷ 4 = 15.

Question 6

(a) 35 — B1

(b) Difference = 50 − 40

= 10 — B1

Tip: half of 100 is 50.